

# Integrating Artificial Intelligence into the Classroom: Evidence from A Conjoint Experiment

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***Abstract- While institutional guidelines for AI use are expanding, there is limited experimental evidence on how students evaluate different forms of AI integration in course design. This study examines how alternative AI policy configurations affect perceived usefulness, support for adoption, and autonomy. Using a conjoint experiment in which students evaluate hypothetical course scenarios varying AI use across instructional materials, grading, and personalized support, we causally estimate the effects of specific design choices. Results show that evaluations are driven primarily by human oversight rather than AI use per se: systems operating without instructor review—especially in grading—produce large declines in all outcomes, whereas AI with oversight is evaluated similarly to fully human instruction. These findings demonstrate that student acceptance of educational AI depends on governance design, identifying human oversight as a key determinant of perceived legitimacy.***

**Keywords - AI adoption, perceived usefulness, perceived autonomy**

## I. INTRODUCTION

Despite rapid integration of artificial intelligence (AI) into higher education, there is limited empirical evidence on how students evaluate different ways in which AI is embedded into course design [1]. In particular, it remains unclear which configurations of AI use, varying in domain and degree of human oversight, students find acceptable, useful, or autonomy-enhancing. [2]. Emerging work on human-AI collaboration suggests that the legitimacy of AI systems depends not only on their technical performance but also on how authority and accountability are allocated between humans and machines [3]. Research in related domains shows that individuals are more willing to accept algorithmic systems when they operate under visible human oversight [4]. Yet, empirical studies have largely focused on students' or instructors' attitudes, leaving open the question of how students evaluate AI when it is embedded into course-level policies.

This study addresses this gap by examining how students evaluate alternative designs of course-level AI policies. Rather than asking whether AI should be used in education, we ask *how* it should be used. We focus on three dimensions central to current debates on AI integration into the classroom [5], AI use in instructional materials, AI use in assessments and grading

[6], and AI-powered personalized support [7], varying each by the degree of automation and human oversight. To do so, we employ a conjoint experiment in which 358 undergraduate students evaluate hypothetical course scenarios [8]. The design mitigates key endogeneity concerns that limit prior observational research, including self-selection into AI-enabled courses and confounding with course or instructor characteristics, allowing for causal estimation of the effects of AI design choices on student perceptions [9].

This study contributes to the literature by providing causal evidence on how the allocation of decision-making authority between AI systems and instructors shapes student evaluations of course design. While prior research has emphasized the importance of human oversight, these claims have largely remained normative or observational. By contrast, this study experimentally shows that student responses vary systematically across pedagogical domains—particularly between instructional and evaluative functions—and that the absence of oversight produces large declines in perceived usefulness, support for adoption, and autonomy, especially in high-stakes contexts such as grading.

## II. CONCEPTUAL FRAMEWORK

This study draws on two related strands of scholarship: research on trust in automation and algorithmic decision-making [10] and technology acceptance and autonomy frameworks [11]. Research on trust in automation emphasizes that users are more willing to rely on AI when human actors retain visible oversight and responsibility for outcomes remains clearly assigned [12]. Fully autonomous systems often generate resistance due to concerns about trust, bias, and loss of control [13]. Research on technology acceptance further suggests that perceived usefulness, support for adoption, and perceived autonomy are central to how individuals evaluate new technologies. These perceptions are also influenced by ethical alignment, transparency, and perceived control [14].

Building on these perspectives, this study advances a conditional framework for AI integration to classroom. Under this framework, students' evaluations of different levels of AI integration in courses depend on the domain in which AI is used and the degree of human oversight retained. AI systems that assist instructors while preserving human review are expected

to be evaluated more favorably than systems that operate without oversight, especially in high-stakes domains such as grading and content generation. By contrast, AI applications that do not directly exercise authority over students, such as personalized support tools, may elicit weaker or neutral responses [15]. This framework motivates the conjoint experiment employed in the study, which decomposes course-level AI policies into 1) instructional materials, 2) assessments and grading, and 3) personalized support, and systematically varies the level of automation and human oversight within each domain. These three domains of AI integration differ not only in their technical function but also in their pedagogical role and the types of learning activities they structure. Instructional materials shape how knowledge is presented and interpreted, positioning AI as a potential source of epistemic authority. Assessments and grading, by contrast, involve evaluative judgment and directly affect students’ academic outcomes, making them high-stakes contexts where concerns about fairness, accountability, and legitimacy are particularly salient. Personalized support tools, such as chatbots, operate in a more optional and student-directed manner, typically assisting with clarification, practice, or navigation of course content.

### III. DATA AND METHODS

The study employs a conjoint experiment to estimate how specific design choices in course-level AI integration influence students’ perceptions of AI use in higher education. Conjoint analysis is well suited to this task because it allows respondents to evaluate multidimensional AI integration scenarios that mirror real-world instructional decision-making, while enabling causal inference [8]. In our experiments, participants evaluated hypothetical course scenarios constructed from a full factorial design crossing three attributes of AI use in education: instructional materials, assessments and grading, and personalized support. Each attribute consisted of three or two levels that varied in the degree of AI automation and human oversight. The full design yielded 18 unique course profiles ( $3 \times 3 \times 2$ ). Profiles were presented in natural language and described how AI was used in each domain within a course. Each participant evaluated six randomly assigned profiles. Table 1 below summarizes the experiment design.

TABLE I: DESIGN OF THE CONJOINT EXPERIMENT

| Attribute               | Levels                                                                                                                                                                                                                     | Conceptual Focus                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Instructional Materials | <ul style="list-style-type: none"> <li>- All materials prepared without AI</li> <li>- Materials generated with AI and reviewed by the instructor</li> <li>- Materials generated by AI without instructor review</li> </ul> | Human oversight in content generation       |
| Assessments & Grading   | <ul style="list-style-type: none"> <li>- Created and graded without AI</li> <li>- AI-assisted quizzes and feedback reviewed by instructor</li> <li>- Generated and graded by AI without review</li> </ul>                  | Automation in evaluation and feedback       |
| Personalized Support    | <ul style="list-style-type: none"> <li>- No AI chatbot support</li> <li>- AI chatbot trained on course content available</li> </ul>                                                                                        | Perceived personalization and accessibility |

After reading each profile, participants rated the course on three continuous outcomes using 0-100 slider scales: 1) perceived usefulness of the AI integration for achieving learning goals, 2) support for adopting similar AI integration in other courses, and 3) perceived autonomy and control over the learning experience. Perceived autonomy was operationalized as students’ reported sense of control over their learning experience and evaluation. It reflects respondents’ perceptions of how much influence they would retain over how they learn, how their work is assessed, and how outcomes are determined. Data were collected in Fall 2025 via an online Qualtrics survey administered through the university of the first author. Undergraduate students were recruited to participate voluntarily and provided informed consent prior to beginning the survey. A total of 363 students initiated the survey, and 358 completed it, yielding a completion rate of 98.6%. After data cleaning, the final analytic sample consisted of 2,148 profile-level observations, with each profile being rated about 120 students. The study protocol was approved by the university’s Institutional Review Board. To estimate the effects of AI integration design choices on student perceptions, we fit linear mixed-effects models separately for each dependent variable. Models included fixed effects for each attribute level and random intercepts for participants to account for within-subject correlation across multiple profile evaluations.

### IV. RESULTS

Fig. 1 summarizes distributional patterns in student evaluations by AI policy configurations. Across instructional materials and grading, AI systems operating without instructor review are associated with substantially lower ratings, while systems with human oversight are rated more favorably. In grading, AI-assisted systems with instructor review closely resemble fully human grading, whereas in instructional materials, reviewed AI performs better than unreviewed AI but remains somewhat lower than fully human preparation. In contrast, AI-powered personalized support shows little systematic effect, suggesting that students perceive such tools as low-stakes or are largely indifferent to their presence.

Figure 1: Distribution of Student Evaluations by AI Course Design

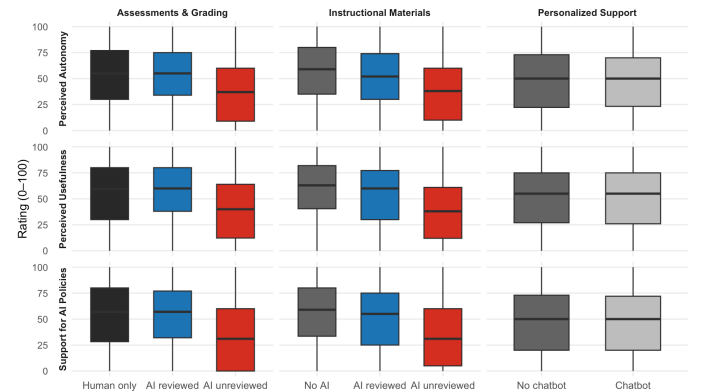


Table 2 reports estimates from linear mixed-effects models. Results mirror Fig. 1. Unreviewed AI in materials produces significant declines across all outcomes ( $\approx -20$  points), while reviewed AI yields smaller but still negative effects ( $\approx -4$  to  $-5$  points), indicating that oversight mitigates but does not eliminate resistance. A similar pattern appears in grading: AI with instructor review does not differ significantly from fully human grading, whereas unreviewed AI leads to substantial declines ( $\approx -16$  points). In contrast, AI chatbot support has no significant effect, showing that students view it as low-stakes or are largely indifferent to its presence.

TABLE II: STUDENT PERCEPTION OF AI INTEGRATION

| Predictor                  | DV1: Usefulness | DV2: Adoption | DV3: Autonomy |
|----------------------------|-----------------|---------------|---------------|
| Materials: AI reviewed     | -5.11 ***       | -5.14 ***     | -4.45 **      |
| Materials: AI unreviewed   | -21.14 ***      | -21.34 ***    | -18.88 ***    |
| Grading: AI reviewed       | +0.44 (ns)      | -0.53 (ns)    | -1.15 (ns)    |
| Grading: AI unreviewed     | -15.12 ***      | -18.33 ***    | -15.87 ***    |
| Support: Chatbot           | -0.16 (ns)      | -0.32 (ns)    | +0.36 (ns)    |
| Conditional R <sup>2</sup> | 0.51            | 0.54          | 0.48          |
| Marginal R <sup>2</sup>    | 0.15            | 0.16          | 0.13          |

Notes: \*\*\*  $p < .001$ , \*\*  $p < .01$ , (ns) = not significant.

Overall, the results demonstrate that systems operating without instructor review reduce perceived usefulness, support for adoption, and autonomy. Although a preference for human oversight may appear expected, these results show that it is conditional rather than uniform, depending on how authority is allocated across course functions, with the strongest resistance arising when AI assumes evaluative roles such as grading.

## V. CONCLUSION

The findings align with theories of trust in automation that emphasize accountability, oversight, and human in the loop [16]. The absence of effects for AI-powered support tools further highlights the importance of role differentiation. When AI functions in an advisory capacity and provides input while leaving the final decision to the human, it may not trigger the same concerns. This interpretation is consistent with prior work showing that AI is often treated differently when it serves as a decision aid rather than making the decision itself [17]. These findings should be interpreted as evidence about student perceptions rather than observed learning outcomes. The single-institution sample and use of hypothetical scenarios limit generalizability, and responses may vary across disciplines, institutional contexts, and task types. Reactions to AI are likely to differ depending on the nature of the activity [18], such as coding versus writing or objective versus subjective evaluation. Future research should examine how these patterns extend to real classroom settings.

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## REFERENCES

- [1] Chan, C. K. Y., & Tsi, L. H. (2024). Will generative AI replace teachers in higher education? A study of teacher and student perceptions. *Studies in Educational Evaluation*, 83, 101395
- [2] Tarisayi, K. S. (2024, March). Strategic leadership for responsible artificial intelligence adoption in higher education. In *CTE workshop proceedings* (Vol. 11, pp. 4-14).
- [3] Wang, D., Churchill, E., Maes, P., Fan, X., Shneiderman, B., Shi, Y., & Wang, Q. (2020, April). From human-human collaboration to Human-AI collaboration: Designing AI systems that can work together with people. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems* (pp. 1-6).
- [4] Holzinger, A., Zatloukal, K., & Müller, H. (2025). Is human oversight to AI systems still possible?. *New Biotechnology*, 85, 59-62.
- [5] Shi, L., Ding, A. C., & Choi, I. (2024). Investigating teachers' use of an AI-enabled system and their perceptions of AI integration in science classrooms: A case study. *Education Sciences*, 14(11), 1187.
- [6] Kumar, R. (2023). Faculty members' use of artificial intelligence to grade student papers: A case of implications. *International Journal for Educational Integrity*, 19(1), 9.
- [7] Bai, Y., & Wang, S. (2025). Impact of generative AI interaction and output quality on university students' learning outcomes: a technology-mediated and motivation-driven approach. *Scientific Reports*, 15(1).
- [8] Giersch, J., & Dong, C. (2018). Principals' preferences when hiring teachers: A conjoint experiment. *Journal of Educational Administration*, 56(4), 429-444.
- [9] Levatino, A., Ferrer-Esteban, G., & Verger, A. (2024). Unveiling teachers' work preferences: A conjoint experiment on the implications of school governance reform across three countries. *Teaching and Teacher Education*, 146, 104631.
- [10] Starke, C., Baleis, J., Keller, B., & Marcinkowski, F. (2022). Fairness perceptions of algorithmic decision-making: A systematic review of the empirical literature. *Big Data & Society*, 9(2), 1-16.
- [11] Xue, L., Mahat, J., & Ghazali, N. (2026). Technology Acceptance Model in Artificial Intelligence in Education: A Meta-Analysis. *Sage Open*, 16(1), 1-23.
- [12] Li, Z. (2024). Generative AI in Higher Education Academic Assignments: Policy Implications from a Systematic Review of Student and Teacher Perceptions (Doctoral dissertation, Massachusetts Institute of Technology).
- [13] Levatino, A., Ferrer-Esteban, G., & Verger, A. (2024). Unveiling teachers' work preferences: A conjoint experiment on the implications of school governance reform across three countries. *Teaching and Teacher Education*, 146, 104631.
- [14] Colonna, L. (2024). Teachers in the loop? An analysis of automatic assessment systems under Article 22 GDPR. *International Data Privacy Law*, 14(1), 3-18.
- [15] Zeiser, J. (2024). Owning decisions: AI decision-support and the attributability-gap. *Science and Engineering Ethics*, 30(4), 27.
- [16] Holzinger, A., Zatloukal, K., & Müller, H. (2025). Is human oversight to AI systems still possible?. *New Biotechnology*, 85, 59-62.
- [17] Bader, V., & Kaiser, S. (2019). Algorithmic decision-making? The user interface and its role for human involvement in decisions supported by artificial intelligence. *Organization*, 26(5), 655-672.
- [18] Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8 (12), 2293-2303.